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Notifications 

3. MATLAB and RREF

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Recitation due Jan 24, 2024 20:30 IST

Reduced row echelon form (RREF) (External resource)

(1.0 points possible)

RREF problem

The steps to calculate the reduced row-echelon form (RREF) of a matrix are quite easy to do them all by hand! Fortunately, MATLAB can calculate the RREF of a matrix for us.

Let A be the following matrix:

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 7 & 4 \\ -3 & 5 & 10 \end{bmatrix}$$

We want to solve the problem:

$$Ax = b$$

with

$$b = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$$

By completing the template below, use RREF to find $x = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$.

Script ?

 Save  Reset

 MATLAB Documentation (<https://www.mathworks.com/help/>)

```
1 % Input your 3x3 matrix A and the 3x1 vector b
2 A = [2 1 1; 1 7 4; -3 5 10];
3 b = [1; -1; 1];
4 % Now create the 3x4 matrix Augmented matrix. You do not need to ty
5 % Think about what we learnt about creating arrays in recitation 1.
6 %
7 Aug = [A,b];
8 %
9 % MATLAB will now calculate the reduced row echelon form of Aug
10 %
11 R = rref(Aug);
12 %
13 % Recall For a 3x3 matrix, the solution x (if it exists) will just
14 % Extract the last column of R and assign it to a variable x.
15 %
16 x = R(:,4)
17 %
18 % Now check that this does indeed solve the problem by calculating
19 b1 = A*x
```

 

Assessment: All Tests Passed

Submit 

 rref present?

 Value of x

 Calculator

✔ check matrix Aug

✔ Check value of b1

Output

3. MATLAB and RREF

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